

Notes: Becko and Sztutman (AER, 2026)

Efficiency Criteria, Income Taxation, and Heterogeneous Elasticities

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1 Big picture

- **Question.** Can economists strengthen the Pareto criterion for income taxation without imposing strong interpersonal comparisons of utility? More concretely, when can an observed nonlinear income tax schedule be justified as utilitarian without relying on extreme cardinalizations of household utility?
- **Classical benchmark.** In a Mirrlees-style nonlinear income tax model, a Pareto-efficient tax schedule can often be rationalized as maximizing utilitarian welfare after choosing suitable cardinal utility scales. This makes Pareto efficiency look closely connected to utilitarian optimality.
- **Main critique.** The rationalization can be normatively fragile. It may require cardinal utility transformations with *unbounded curvature* in consumption. Economically, the planner's marginal willingness to transfer resources to some households must fall arbitrarily quickly as their consumption changes.
- **New welfare criterion.** Becko and Sztutman ask whether taxes can be rationalized by utilitarian preferences with **bounded curvature**. This is a small strengthening of Pareto efficiency: it does not pick specific welfare weights, but it rules out cardinalizations with arbitrarily extreme local curvature.
- **Theoretical result.** A tax schedule can be rationalized with bounded curvature if and only if revenues are both **decreasing** and **concave** with respect to a class of narrowly targeted, two-bracket tax reforms.
- **Why concavity matters.** Pareto efficiency only asks whether a small targeted reform raises revenue to first order. Bounded-curvature rationalizability also asks whether repeating the same reform raises revenue more or less than the first time. If revenue is locally convex, the planner can exploit the convexity through reforms that sort taxpayers by elasticity.
- **Mechanism.** The central force is **within-income heterogeneity in elasticities of taxable income**. A targeted tax cut can pull high-elasticity households into the bracket where marginal taxes are cut and push low-elasticity households into the bracket where marginal taxes rise. The paper calls the logic a “sort-and-extort” mechanism.
- **Empirical test.** The paper reformulates the theoretical condition as a sufficient-statistics test using the tax schedule, the income-density Pareto parameter, conditional mean ETIs, and conditional ETI variance.
- **Main empirical result.** Using a panel of US tax returns, the paper estimates conservative lower bounds on ETI variance conditional on income. The 1990 US tax schedule appears likely Pareto efficient but close to, and plausibly beyond, the boundary for rationalizability with bounded curvature.
- **Quantitative implication.** Under an illustrative calibration, welfare gains can arise from either lowering the top marginal tax rate to about **13.3%** or raising it to about **71.2%**. The two directions are not a contradiction: the point is that the current tax schedule violates a second-order condition, so some reforms that change marginal incentives among top earners can improve welfare.
- **Takeaway.** The paper offers a way to discipline inverse-optimum arguments. It is not enough that observed taxes can be rationalized by some welfare weights; the cardinalization may be too extreme if revenues are locally convex in targeted reforms.

2 Model and objects (key definitions)

2.1 Environment

- Time is static. There is a unit measure of households $h \in \mathcal{H}$ and a representative competitive firm.
- Output is produced one-for-one from labor, so the wage is normalized to one.
- The government chooses a nonlinear **income retention schedule**

$$R : \mathbb{R}_+ \rightarrow \mathbb{R}_+,$$

where $R(z)$ is posttax income when pretax income is z . Equivalently, the tax liability is

$$T(z) = z - R(z).$$

- Given R , household h chooses income/labor supply z_h and consumption c_h by solving

$$\max_{c_h, z_h} u_h(c_h, z_h) \quad \text{subject to} \quad c_h = R(z_h). \quad (1)$$

- When the solution is unique, write the induced choices and indirect utility as

$$c_h(R), \quad z_h(R), \quad V_h(R).$$

- Government net revenue is

$$\text{Rev}(R) = \mathbb{E}_h[z_h(R) - c_h(R)]. \quad (2)$$

Interpretation. The paper is not solving a standard optimal tax problem with fixed welfare weights. It starts from an observed or baseline tax schedule R_0 and asks whether small reforms can improve it under different welfare criteria.

2.2 Local tax reforms

The paper studies paths of nearby retention schedules

$$R(\iota) = R_0 + \iota\Delta + \frac{\iota^2}{2}\Gamma, \quad (3)$$

where Δ is the first-order direction of reform and Γ is the second-order component.

A key reform direction is a narrowly targeted bump in retention around income z :

$$\Delta^*(\hat{z}; z, \ell) = \tilde{\Delta} \left(\frac{\hat{z} - z}{\ell} \right),$$

where $\tilde{\Delta}$ is a smooth bump that equals one at zero and zero outside $(-1, 1)$.

Interpretation. This is a smooth version of a two-bracket reform. It raises retention locally around income z and leaves taxes unchanged outside a small interval. Its slope changes marginal tax rates in a narrow neighborhood.

2.3 Revenue derivatives for targeted reforms

The two central objects are the first and second derivatives of revenue with respect to the targeted reform:

$$\delta\text{Rev}(z) \equiv \lim_{\ell \rightarrow 0^+} \frac{d}{d\ell} \text{Rev}(R_0 + \iota \Delta^*(z, \ell)) \Big|_{\iota=0}, \quad (4)$$

$$\delta^2\text{Rev}(z) \equiv \lim_{\ell \rightarrow 0^+} \frac{d^2}{d\ell^2} \text{Rev}(R_0 + \iota \Delta^*(z, \ell)) \Big|_{\iota=0}. \quad (5)$$

- $\delta\text{Rev}(z)$ is the **first-order revenue effect** of a narrow tax cut at income z .
- $\delta^2\text{Rev}(z)$ is the **curvature of the revenue function** in that targeted direction.

Sign convention. Since Δ^* raises retention, it is a local tax cut. Thus:

- $\delta\text{Rev}(z) > 0$ means the targeted tax cut raises revenue to first order.
- $\delta\text{Rev}(z) < 0$ means the targeted tax cut loses revenue to first order.
- $\delta^2\text{Rev}(z) > 0$ means revenue is locally convex in the targeted reform.

3 Welfare criteria and theory (all equations + interpretation)

3.1 Three welfare notions

The paper compares three notions of local efficiency.

1. **Deterministic Pareto efficiency.** A reform that makes someone better off and no one worse off must lower government revenue. If a Pareto improvement also raises revenue, then the original tax schedule is locally inefficient.
2. **Random Pareto efficiency.** The same idea, but allowing random tax schedules. A random reform is attractive if it improves expected utility for all households and raises expected revenue.
3. **Utilitarian rationalizability.** The tax schedule R_0 is locally rationalizable if there exist strictly increasing cardinalizing functions w_h such that R_0 locally maximizes

$$W(R) = \mathbb{E}_h [w_h(V_h(R))] \quad (6)$$

subject to the revenue requirement.

Interpretation. Pareto efficiency uses only ordinal preferences. Utilitarian rationalizability chooses cardinal scales for utility and then asks whether the observed tax schedule is locally optimal for some utilitarian planner.

3.2 First-order result: Pareto efficiency

The first key result links Pareto efficiency to the sign of $\delta\text{Rev}(z)$.

- If there is some income z such that

$$\delta \text{Rev}(z) > 0,$$

then the targeted local tax cut raises revenue and improves all directly affected households. The tax schedule is locally Pareto inefficient and cannot be rationalized by any increasing cardinalization.

- If, under regularity conditions,

$$\delta \text{Rev}(z) < 0 \quad \text{for all } z > 0,$$

then the tax schedule is locally Pareto efficient and can be rationalized by some continuously differentiable cardinalization.

Interpretation. Pareto efficiency is a first-order condition. It rules out local reforms that mechanically create a Pareto improvement and finance themselves. It does not rule out extreme welfare rationalizations.

3.3 Bounded-curvature rationalizability

The paper's central normative restriction is that the cardinalizations w_h should not make marginal social values change infinitely fast around the observed allocation. A simplified version of the bounded-curvature requirement is

$$c_{h0} \left| \frac{\partial^2 (w_h \circ u_h)(c_{h0}, z_{h0}) / \partial c^2}{\partial (w_h \circ u_h)(c_{h0}, z_{h0}) / \partial c} \right| \leq \bar{\rho}(z_{h0}) \quad \text{for all } h, \quad (7)$$

for some finite function $\bar{\rho}(\cdot)$.

Interpretation.

- Ex post: social marginal values of transfers cannot change arbitrarily quickly with small consumption changes.
- Ex ante: behind a veil of ignorance, the planner's implied risk aversion is bounded.
- This still permits very high curvature; it only rules out unbounded curvature.

3.4 Second-order result: revenue concavity

The second key result is that bounded-curvature rationalizability requires a second-order revenue condition.

- If there exists an income z such that

$$\delta^2 \text{Rev}(z) > 0,$$

then no locally bounded-curvature cardinalization can rationalize the tax schedule.

- Conversely, under regularity conditions, if

$$\delta \text{Rev}(z) < 0 \quad \text{and} \quad \delta^2 \text{Rev}(z) < 0 \quad \text{for all } z > 0,$$

then there exist bounded-curvature cardinalizations that locally rationalize R_0 .

Interpretation. Pareto efficiency asks whether targeted tax cuts lose revenue. Bounded-curvature rationalizability asks more: targeted tax cuts must also face diminishing returns. If revenue is convex, a planner can exploit that convexity with a nearby reform.

3.5 Random versus deterministic intuition

The random-reform argument is the cleanest way to see why convexity matters.

- Suppose a random reform raises retention in one state and lowers retention in another.
- If revenue is convex in the reform, expected revenue rises even when the average reform is zero.
- Risk aversion can make households dislike the randomization, but if the reform is narrow enough, second-order revenue gains dominate bounded utility-curvature costs.

The deterministic counterpart places the tax increase and tax decrease in nearby income brackets rather than across states of the world.

Interpretation. Convex revenue creates room for a reform that is budget-improving to second order. Bounded curvature prevents the planner from assigning infinite disutility to the households who lose from the deterministic version.

4 Sufficient-statistics formulas

4.1 Elasticity concepts

The empirical content comes from rewriting $\delta\text{Rev}(z)$ and $\delta^2\text{Rev}(z)$ using local elasticities.

- η_h : income elasticity of taxable income with respect to the level of retention.
- ϵ_h : compensated elasticity of taxable income with respect to marginal retention.
- ϵ_h^+ : compensated super-elasticity, capturing how ϵ_h changes when income changes.

Income-conditional moments are

$$\eta(z) = \mathbb{E}[\eta_h \mid z_{h0} = z], \quad (8)$$

$$\epsilon(z) = \mathbb{E}[\epsilon_h \mid z_{h0} = z], \quad (9)$$

$$m_2(z) = \mathbb{E}[\epsilon_h^2 \mid z_{h0} = z], \quad (10)$$

$$m_+(z) = \mathbb{E}[\epsilon_h \epsilon_h^+ \mid z_{h0} = z]. \quad (11)$$

The local Pareto parameter of the income distribution is

$$\alpha(z) = -\frac{d \log(zg(z))}{d \log z}, \quad (12)$$

where $g(z)$ is the income density.

Interpretation. $\alpha(z)$ measures how fast the income density thins out around z . At high incomes, the density is steeply falling, which amplifies sorting effects.

4.2 First-order revenue effect

The paper shows that

$$\begin{aligned} \delta \text{Rev}(z) = \kappa_1(z) & \left[-1 + (1 - R'_0(z)) \frac{z\eta(z)}{R_0(z)} \right. \\ & \left. + \left\{ (1 - R'_0(z)) \left(\alpha(z) - \frac{d \log \epsilon(z)}{d \log z} \right) + \frac{R''_0(z)z}{R'_0(z)} \right\} \frac{\epsilon(z)}{R'_0(z)} \right], \end{aligned} \quad (13)$$

where $\kappa_1(z) > 0$.

Interpretation.

- The -1 term is the mechanical revenue loss from giving households more retention.
- The $\eta(z)$ term reflects income effects.
- The terms involving $\alpha(z)$, $\epsilon(z)$, and the tax schedule capture behavioral fiscal externalities from changes in marginal tax rates.

4.3 Second-order revenue effect

The second-order revenue effect is

$$\delta^2 \text{Rev}(z) = \kappa_2(z) \left[-(1 + R'_0(z))\epsilon(z) + (1 - R'_0(z)) \left(\alpha(z)m_2(z) - \frac{dm_2(z)}{d \log z} + m_+(z) \right) \right], \quad (14)$$

where $\kappa_2(z) > 0$.

Interpretation.

- The negative term $-(1 + R'_0)\epsilon$ pushes toward revenue concavity.
- The positive term $(1 - R'_0)\alpha m_2$ pushes toward revenue convexity.
- The appearance of the second moment $m_2(z) = \mathbb{E}[\epsilon_h^2 \mid z]$ is the key. High within-income ETI variance can make revenue locally convex.

4.4 Top-income diagnostic

At top incomes, with no income effects and no super-elasticities, let

$$\tau_{top} = 1 - R'_0(z), \quad \alpha_{top} = \lim_{z \rightarrow \infty} \alpha(z), \quad \epsilon_{top} = \lim_{z \rightarrow \infty} \epsilon(z),$$

and let $\sigma_{top}^2 = \text{Var}(\epsilon_h \mid z_{h0} = z)$ at the top.

The first-order Pareto-efficiency test fails if

$$\tau_{top} > \frac{1}{1 + \alpha_{top}\epsilon_{top}}. \quad (15)$$

The second-order bounded-curvature test fails if

$$\tau_{top} > \frac{2}{1 + \alpha_{top} \left(\epsilon_{top} + \sigma_{top}^2 / \epsilon_{top} \right)}. \quad (16)$$

Thus bounded-curvature rationalizability requires

$$\tau_{top} < \min \left\{ \frac{1}{1 + \alpha_{top}\epsilon_{top}}, \frac{2}{1 + \alpha_{top}(\epsilon_{top} + \sigma_{top}^2/\epsilon_{top})} \right\}. \quad (17)$$

Key implication. A tax schedule can pass the first-order Pareto test but fail the second-order bounded-curvature test when σ_{top}^2 is large enough.

5 Mechanism: heterogeneity and sorting

5.1 Why ETI variance creates revenue convexity

Consider a local reform that lowers marginal taxes around income z .

- If the income density is steeply declining, there are more taxpayers just below z than just above z .
- A tax cut around z can pull more taxpayers into the bracket from below than it pushes out above.
- High-elasticity households move farther in response to the reform.
- Therefore, the reform tends to concentrate high-elasticity households exactly where marginal taxes fall.
- Repeating the reform then has a larger positive behavioral revenue effect than the first reform.

Interpretation. This is the paper’s “sort-and-extort” mechanism: the tax system sorts households by elasticity across nearby incomes, then exploits the sorting by changing marginal taxes differently across those nearby incomes.

5.2 Why one-dimensional heterogeneity is not enough

The paper shows that, in a standard class of preferences with a single group,

$$u_h(c, z) = v^c(c) - \theta_h v^z(z), \quad (18)$$

first-order and second-order revenue effects move together. In this case,

$$\delta \text{Rev}_i(z) < 0 \quad \Rightarrow \quad \delta^2 \text{Rev}_i(z) < 0. \quad (19)$$

Interpretation. If heterogeneity is essentially one-dimensional, the tax planner cannot use small income-bracket reforms to change the local elasticity composition in a way that generates convex revenue. Failures of bounded-curvature rationalizability require richer heterogeneity within income levels.

5.3 Economic meaning of failure

If taxes are Pareto efficient but fail the second-order test:

- no simple first-order Pareto-improving tax cut exists;
- nevertheless, some nearby reform can improve utilitarian welfare for every bounded-curvature cardinalization;
- the rationalization of the observed schedule relies on extreme, unbounded curvature in utility transformations.

6 Empirical test and identification

6.1 Data

- The empirical analysis uses the NBER panel of US tax returns and related tax-return data.
- The key period is 1979–1990, a period with large federal tax reforms and variation in state tax schedules.
- The empirical target is the shape of tax revenue around the US income tax schedule, especially near the top of the income distribution.

Why top incomes? The second-order test is most likely to fail where marginal tax rates, mean elasticities, ETI variance, and the Pareto shape parameter are high.

6.2 Mean ETI by income

The paper estimates mean ETIs nonparametrically by income level using a local regression similar to Gruber and Saez (2002):

$$\Delta \log z_{ht} = a(\tilde{z}) + \epsilon(\tilde{z}) \Delta \log R'_t(z_{h,t-1}) + c(\tilde{z}) mrs_{h,t-1} + d_{t-1} + \xi_{ht}. \quad (20)$$

- $\tilde{z}$ is year-demeaned log income.
- $\Delta \log R'_t(z_{h,t-1})$ is the change in marginal retention evaluated at the household's initial income.
- Controls include marital status and time fixed effects.

Identification. Conditional on controls and income level, changes in marginal tax rates are assumed to be as good as randomly assigned. Variation comes from federal reforms, state reforms, and differential changes across income levels.

6.3 Lower bound on ETI variance

The main empirical challenge is to estimate

$$\text{Var}(\epsilon_h \mid z_{h0} = z),$$

not merely average ETIs.

The paper uses itemization behavior to construct observable groups with different elasticities. At each income level, households are classified as heavy or light itemizers based on total deductions. The regression is

$$\begin{aligned}\Delta \log z_{ht} = & a(\tilde{z}) + \epsilon_{L,t}(\tilde{z}) \Delta \log R'_t(z_{h,t-1}) \\ & + \delta(\tilde{z}) HI_{h,t-1} \Delta \log R'_t(z_{h,t-1}) + c(\tilde{z}) mrs_{h,t-1} + d_{t-1} + \xi_{ht}.\end{aligned}\tag{21}$$

- $HI_{h,t-1}$ indicates heavy itemizers.
- $\delta(\tilde{z})$ measures the within-income difference in elasticity between heavy and light itemizers.
- Differences in group means imply a lower bound on total within-income variance:

$$\text{Var}(\epsilon_h \mid z) \geq p(z)(1 - p(z))(\epsilon_H(z) - \epsilon_L(z))^2.$$

Why this is conservative. The lower bound uses only between-group variation. Any unobserved variation within itemizer groups would make the true ETI variance larger.

6.4 Main empirical findings

- Mean ETIs are typically between **0.2 and 0.3** across most incomes.
- Light itemizers have elasticities near zero, while heavy itemizers have elasticities near one.
- The two-group itemization split implies a lower bound on ETI variance of about **0.2** at many income levels.
- Further subdividing high-elasticity groups gives a top-income lower bound around **0.25**.
- With $\alpha \approx 2.5$, $\epsilon \approx 0.3$, and $\tau_{top} \approx 50\%$, the first-order test is satisfied, but the second-order test is very close to failure.
- The estimated top ETI standard deviation lower bound is $\hat{\sigma}_{top} = \sqrt{0.25} = 0.5$. If the true standard deviation is more than about **4%** above that lower bound, the second-order condition fails.

Interpretation. The US schedule in the sample period looks Pareto efficient in the first-order sense but likely cannot be rationalized by bounded-curvature utilitarian preferences once realistic ETI heterogeneity is allowed.

6.5 What makes the estimates credible

- The estimation strategy uses tax variation induced by policy changes rather than cross-sectional correlations alone.
- The ETI variance estimate is explicitly a lower bound, making the test conservative.
- Splitting itemizers into more groups sharpens the lower bound and produces similar implications at top incomes.
- The test is framed in sufficient statistics: the paper does not need to estimate the full distribution of preferences or solve the complete Mirrlees problem.

6.6 Important limitations

- The empirical test focuses on intensive-margin taxable income responses.
- The local empirical implementation uses simplifying assumptions: no income effects, CES labor disutility, and medium-run uncompensated ETIs as the relevant compensated elasticities.
- Itemization-based heterogeneity may not capture all relevant ETI heterogeneity. This is why the paper treats it as a lower bound rather than the full variance.
- The welfare criterion is local. It diagnoses whether nearby reforms can improve the schedule; it does not by itself deliver a full global optimal tax schedule.

7 Quantitative application

7.1 Calibration

The paper illustrates the magnitude of possible welfare gains with a simple top-income model.

- Focus on households with income above \$100,000 in 1990 dollars.
- Top income distribution: Pareto with shape $\alpha_{top} = 2.5$.
- Initial top marginal tax rate: $\tau_{top} = 50\%$.
- ETI distribution: truncated log-normal with mean 0.3, maximum 12, and standard deviation 0.75.
- Planner places a common welfare value on transfers to top earners. Matching the first-order condition implies

$$\lambda_{top} = 0.25.$$

Interpretation. The calibration chooses welfare weights so that the planner is consistent with the observed tax rate at first order. The exercise then asks whether the schedule fails at second order.

7.2 Welfare gains from simple top-tax reforms

Under the calibration, the second-order condition is violated at top incomes. The paper then considers simple reforms of the top marginal tax rate.

- Lowering the top marginal tax rate to **13.3%** generates welfare equivalent to about **\$1,966 per top earner**.
- Raising the top marginal tax rate to **71.2%** generates welfare equivalent to about **\$972 per top earner**.
- The revenue-maximizing top tax rate is about **80.9%** in the baseline calibration.

Interpretation. The possibility of gains in both directions reflects local nonconcavity. The current schedule is not locally pinned down by a well-behaved second-order condition, so different reforms can exploit the distribution of elasticities in different ways.

8 Figures

Figure 1 and Figure 2: Local and reform-comparison diagrams

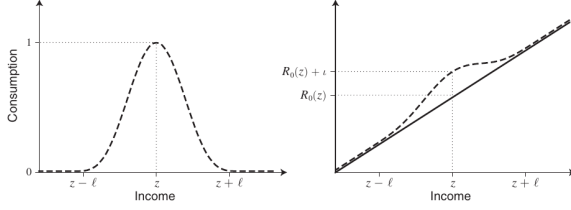


Figure 1: Local Reform to an Income Retention Schedule

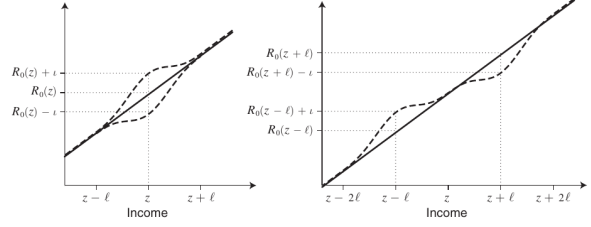


Figure 2: Random and Deterministic Reforms

Read: Local Reform to an Income Retention Schedule.

- The left panel shows the smooth local reform direction $\Delta^*(z, \ell)$.
- The right panel shows how adding a small multiple of this reform changes the initial retention schedule R_0 .
- The point of the figure is that the paper studies extremely local changes in marginal incentives while keeping the tax schedule smooth.

Read: Random and Deterministic Reforms.

- The left panel shows the random-reform idea: retention is raised or lowered around the same income in different states.
- The right panel shows the deterministic analog: taxes are changed in adjacent income brackets.
- The figure explains how random-reform convexity intuition becomes a deterministic welfare-improving reform when bounded-curvature cardinalizations are imposed.

Figure 3: Empirical diagnostic diagram

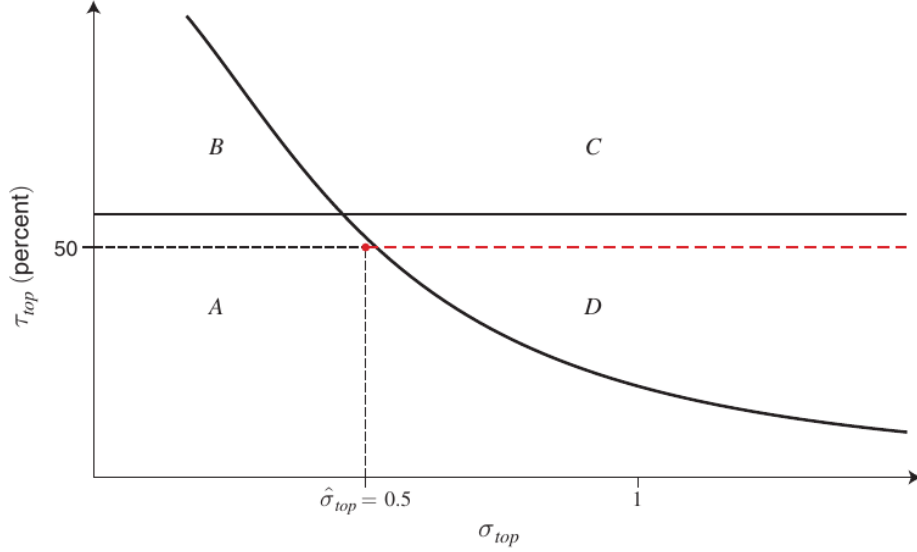


Figure 3: Visualization of First- and Second-Order Tests

Read: Visualization of First- and Second-Order Tests.

- The vertical axis is the top marginal tax rate τ_{top} .
- The horizontal axis is the top-income ETI standard deviation σ_{top} .
- Region A: neither test is violated.
- Region B: only the first-order Pareto test is violated.
- Region C: both tests are violated.
- Region D: only the second-order bounded-curvature test is violated.
- The dashed horizontal line is $\tau_{top} = 50\%$.
- The dashed vertical line is the estimated lower bound $\hat{\sigma}_{top} = 0.5$.

Main message. The empirical estimate places the US top tax schedule very close to the boundary between region A and region D. Since the standard deviation estimate is only a lower bound, the paper argues that the second-order condition is likely to fail.

9 Short summary

- **Problem.** Pareto-efficient taxes can be rationalized by utilitarian welfare only by choosing cardinal utility scales. Some rationalizations are extreme.
- **Criterion.** Require the cardinalizations to have bounded curvature in consumption.
- **Theory.** Pareto efficiency is equivalent to $\delta\text{Rev}(z) < 0$ for all z . Bounded-curvature rationalizability requires both $\delta\text{Rev}(z) < 0$ and $\delta^2\text{Rev}(z) < 0$ for all z .

- **Mechanism.** High within-income ETI variance makes revenues convex because tax reforms sort high- and low-elasticity households across nearby income brackets.
- **Empirics.** US tax return data imply sizable lower bounds on ETI variance. The 1990 tax schedule appears to satisfy the first-order test but is very close to failing, and plausibly fails, the second-order test.
- **Policy implication.** The observed schedule may be Pareto efficient but still inconsistent with any non-extreme utilitarian rationalization. Nearby reforms can generate welfare gains under bounded-curvature welfare criteria.

10 Conclusion

Core conclusion. Becko and Sztutman provide a new efficiency criterion that lies between Pareto efficiency and fully specified utilitarianism. It asks whether an observed income tax schedule can be rationalized by utilitarian welfare without unbounded curvature in utility cardinalizations.

Economic intuition. Pareto efficiency checks only first-order revenue effects. Bounded-curvature rationalizability also requires local revenue concavity. When ETIs are heterogeneous within income levels, narrowly targeted reforms can sort taxpayers by elasticity and generate revenue convexity. That convexity destroys well-behaved utilitarian rationalizability even when Pareto efficiency survives.

Takeaway. The paper turns the abstract issue of interpersonal utility cardinalization into an estimable tax test. The critical sufficient statistic is not just the mean ETI but the variance of ETIs among taxpayers with the same income.